MILITARY SPECIFICATION
TEST SET, MISSION RECORDER AN/USM-387

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for ground support intermediate (shop) maintenance of the Mission Recorder Set AN/USH-17(V). The equipment shall generate test signals for the record electronics and monitor reproduced signals of the Mission Recorder Set.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Drawing or Model Number	Nomenclature	Reference Paragraph
OP-87/USM-387	Power Supply Group	3.7.1
OQ-184/USM-387	Test Set Group, Recorder	3.7.2
MT-4628/USM-387	Test Stand, Recorder	3.7.3
	Tape, Recorder Test	3.7.4

1.3 Associated equipment - The equipment shall operate with the following associated equipment:

Multimeter, VOM, AN/USM-311 (FSN 2Z6625-160-1301)

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue listed or of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 5841

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SPECIFICATIONS

Military

MIL-J-641/10	Jack, Telephone, Type JJ-083
MIL-C-3767/13	Connector, Plug, Electrical (Power, Three-Wire, Fixed Grounding Type) Type UP221M
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-16400F 24 February 1966 and Amendment 4 15 May 1968	Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-21097/14	Connector, Electrical, Male Adapter, Printed Wiring, Contact Spacing, 0.200, Alternate Dual Row, Type CS, Blade Contact (Mounting Provision A)
MIL-T-21200	Test Equipment for Use With Electronic and Electrical Equipment, General Specification for
MIL-C-39012/21	Connector, Coaxial, Radio Frequency (Series Bnc (Uncabled) - Receptacles, Female, Jam Nut Mounted, Class II)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment

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MIL-STD-461A 1 August 1968 and Notice 1 7 February 1969	Electronic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-749	Preparation and Submission of Data for Approval of Nonstandard Electronic Parts
MIL-STD-761B 15 July 1965	Electric Power, Alternating Current for Shipboard Use, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS3106	Connector, Plug, Electric, Straight
MS3120	Connector, Receptacle, Electric, Crimp-type, Wall Mounting, Flange, No. 4 Holes, Bayonet Coupling
MS17325	Meter, Time Totalizing, Digital, 115 Volt, 60 or 400 Hertz
MS24264	Connector, Receptacle, Electrical Flange Mount, Miniature
MS24266	Connector, Plug, Electrical-Straight, Miniature

Naval Air Systems Command

AR-10	Maintainability of Avionic Equipment and Systems, General Requirements for

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2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Testing - This specification makes provision for testing (see 4).

3.2 Materials, parts, and processes - The materials, parts and processes shall be in accordance with MIL-T-21200 or MIL-E-16400 (where specified).

3.2.1 Selection of parts, materials, and processes - Except as specified herein, MIL-T-21200 or MIL-E-16400 (where specified) shall apply for the selection of parts, materials and processes.

3.2.2 Approval of nonstandard parts and materials - Approval for the use of nonstandard parts and materials shall be in accordance with Step I and Step II only of MIL-STD-749.

3.2.3 Samples required for parts and material approvals - When specified by the procuring activity, samples of nonstandard parts shall be submitted in accordance with MIL-T-21200.

3.3 Design and construction - The design and construction of the equipment shall be in accordance with MIL-T-21200 or MIL-E-16400 (where specified).

3.3.1 Total weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 460 pounds.

3.3.2 Microelectronic items - Microelectronic items shall be hermetically sealed.

3.3.3 Electron devices - Electron tubes and semiconductor devices shall be in accordance with MIL-T-21200.

3.3.4 Modules, maintenance - The electronic portions of the equipment shall be divided into maintenance modules as specified in AR-10. Modules shall be considered repairable unless otherwise specified.

3.3.5 Headset jacks - Headset jacks shall be Type M641/10-01, in accordance with MIL-J-641.

3.3.6 Time totalizing meter - The equipment shall contain time totalizing meters conforming to MS17325. A time totalizing meter shall be included in the following units.

Unit	Type of Meter
Power Supply	MS17325-2
Radar Indicator	MS17325-2
Signal Generator	MS17325-2

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3.3.7 Cables and connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200 or MIL-E-16400 (where specified). Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.7.1 Cabling - The test equipment cabling shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested.

3.3.7.2 External connectors - External connectors shall provide for the following (see figure 9):

Reference Designation	Connector Type	Function
2A3J1	M39012/21-0002	Reproduce Digital Data Output
2A3J2	M39012/21-0002	Reproduce Digital Data Sync Output
2A3J3	M39012/21-0002	Reproduce Test Sync Output
2A3J4	M641/10-01	External Audio Input
2A3J7	M39012/21-0002	TV Monitor Video Output
W1P1	MIL-C-3767/13 Type UP221M	Power Input to Power Supply Group
W7P2	MS24266G24-B57SN	Record Signal Output to Mission Recorder Electronics Unit MX-9226(V)/USH-17(V)
W7P3	MS24266G18 - B31S6	Power Output and Signal Connection for Recorder Remote Control Unit C-9071/USH-17(V)
W7P4	MS3106F-16-11P	+28 VDC Power Input
W8P1	MS3106F-20-4P	115 VAC, 400 Hz Power Input

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Reference Designation	Connector Type	Function
W8P2	MS24266G12B3S6	115 VAC, 400 Hz Power Output to the Mission Recorder Electronics Unit, MX-9226(V)/USH-17(V)
W9P1	MS3106F-16-11P	+28 VDC Power Input
W9P2	MS24266G14B4SN	+28 VDC Power Output to the Mission Recorder Tape Transport Unit, RO-429(V)/USH-17(V)
W10P1	MS24266G22B32PN	Signal Connection for Mission Recorder Electronics Unit, MX-9226(V)/USH-17(V)
W10P2	MS24266G22B32SN	Signal Connection for Mission Recorder Tape Transport Unit, RO-429(V)/USH-17(V)
W11P1	MS24266G24B43PN	Power Input and Control Signal Connection from Mission Recorder Electronics Unit, MX-9226(V)/USH-17(V)
W11P2	MS24266G24B43SN	Power Output and Control Signal Connection to Mission Recorder Tape Transport Unit, RO-429(V)/USH-17(V)
W12P2	MS24266G20B28PN	Reproduce Signal Input from Mission Recorder Electronics Unit MX-9226(V)/USH-17(V)

3.3.8 Controls - The front panel controls shall be in accordance with MIL-T-21200. Controls shall provide for the following:

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Control	Function
RADAR SCAN Switch	Signal Generator control which selects ARE or PPI scan radar channel signals for the Mission Recorder and Radar Indicator.
ARE LEVEL Switch	Signal Generator control which selects ARE 30 or ARE 60 digital channel test signals for the Mission Recorder.
FUNCTION Switch	Signal Generator control which selects SELF TEST, RECORD or REPRODUCE testing function of the Test Set.
ATTACK COMMAND Switch	Signal Generator control which switches simulated attack signal ON or OFF.
LAMP TEST Switch	Signal Generator control which, when pressed, verifies operation of FAULT and RADAR MODE indicator lamps.
RADAR MODE Switch	Signal Generator control (located behind front panel) which selects simulated AN/APQ-92 or AN/APQ-148 radar signals for the Mission Recorder.
AUDIO LEVEL Control	Radar Indicator control used to adjust speaker audio level.
CONTRAST Control	Radar Indicator control used to adjust CRT display contrast.
BRIGHTNESS Control	Radar Indicator control used to adjust CRT display brightness.
HORIZONTAL Control	Radar Indicator control used to adjust horizontal positioning of the CRT display.
VERTICAL Control	Radar Indicator control used to adjust vertical positioning of the CRT display.

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Control	Function
115 VAC Switch	Power Supply control used to switch 115 VAC, 60 Hz power for the equipment ON or OFF.
+32 VDC Circuit Breaker	Power Supply pushbutton reset circuit breaker used to provide overload protection for 115 VAC, 60 Hz power input to the +32 VDC power supply subassembly.
-32 VDC Circuit Breaker	Power Supply pushbutton reset circuit breaker used to provide overload protection for 115 VAC, 60 Hz power input to the -32 VDC power supply subassembly.
BLOWER Circuit Breaker	Power Supply pushbutton reset circuit breaker used to provide overload protection for 115 VAC, 60 Hz power to the equipment (except +32 VDC and -32 VDC power supply subassemblies).

3.3.9 Human engineering - Human engineering shall be in accordance with MIL-STD-454, Requirement 62.

3.4 Availability -

3.4.1 Reliability - When specified by the procuring activity, the contractor shall conduct a reliability program in accordance with Requirement 35 of MIL-STD-454. The program plan shall be approved by the procuring activity.

3.4.2 Maintainability - When specified by the procuring activity, the contractor shall conduct a maintainability program in accordance with Requirement 54 of MIL-STD-454. The program plan shall be approved by the procuring activity.

3.4.3 Interference control - The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be controlled within the limits of MIL-STD-461 for class 1C equipment except for the following:

(1) Power lead conducted emissions - When tested in accordance with Test Method CE01, the conducted emissions shall not exceed the limits by more than 20 dB.

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(2) Magnetic field radiated emissions - When tested in accordance with Test Method RE01, magnetic interference measurements shall be made at a distance of 24 inches, minimum, from the face of the Radar Indicator CRT.

3.4.4 Maintenance provisions and field testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.4.5 Nomenclature and nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.6 Service conditions - Service conditions (environmental) shall be in accordance with MIL-T-21200 for class 3 equipment.

3.4.7 Standard conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input power voltage	115 VAC, 60 Hz

3.5 Transportability - Except as specified herein, transportability requirements shall be in accordance with MIL-T-21200.

3.5.1 Forklift provisions - The equipment shall not be designed for lifting or transporting by forklift vehicles.

3.6 Performance characteristics - The performance of the equipment shall be as specified herein.

3.6.1 Output signals - The equipment shall provide the output signals described herein to the Mission Recorder. Timing of output signals shall be as shown in referenced figures.

3.6.1:1 Horizontal Sweep output - The output signal shall conform to Waveform A, Figure 1, 2, 3 or 4; depending upon selected mode and scan.

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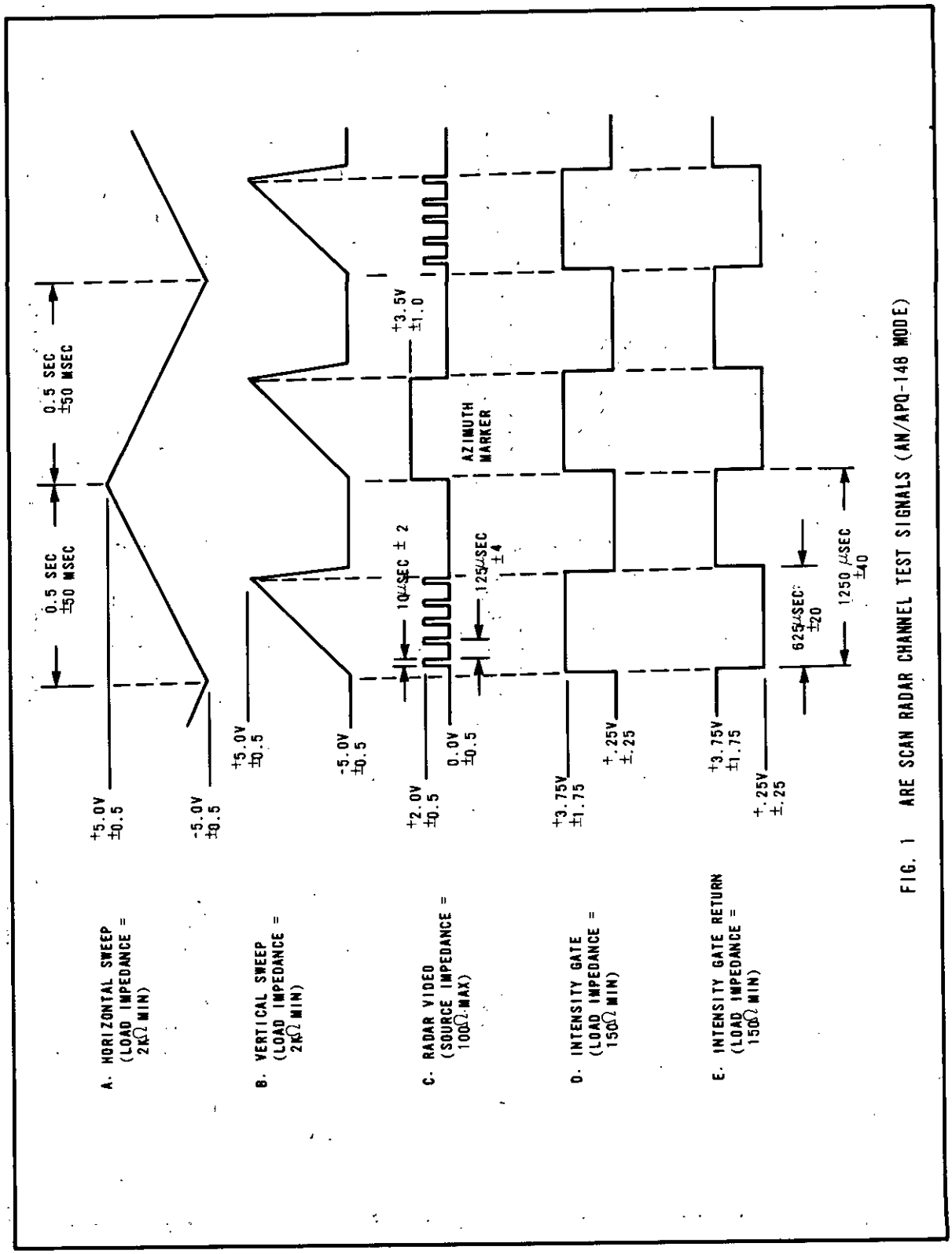


FIG. 1 ARE SCAN RADAR CHANNEL TEST SIGNALS (AN/APQ-148 MODE)

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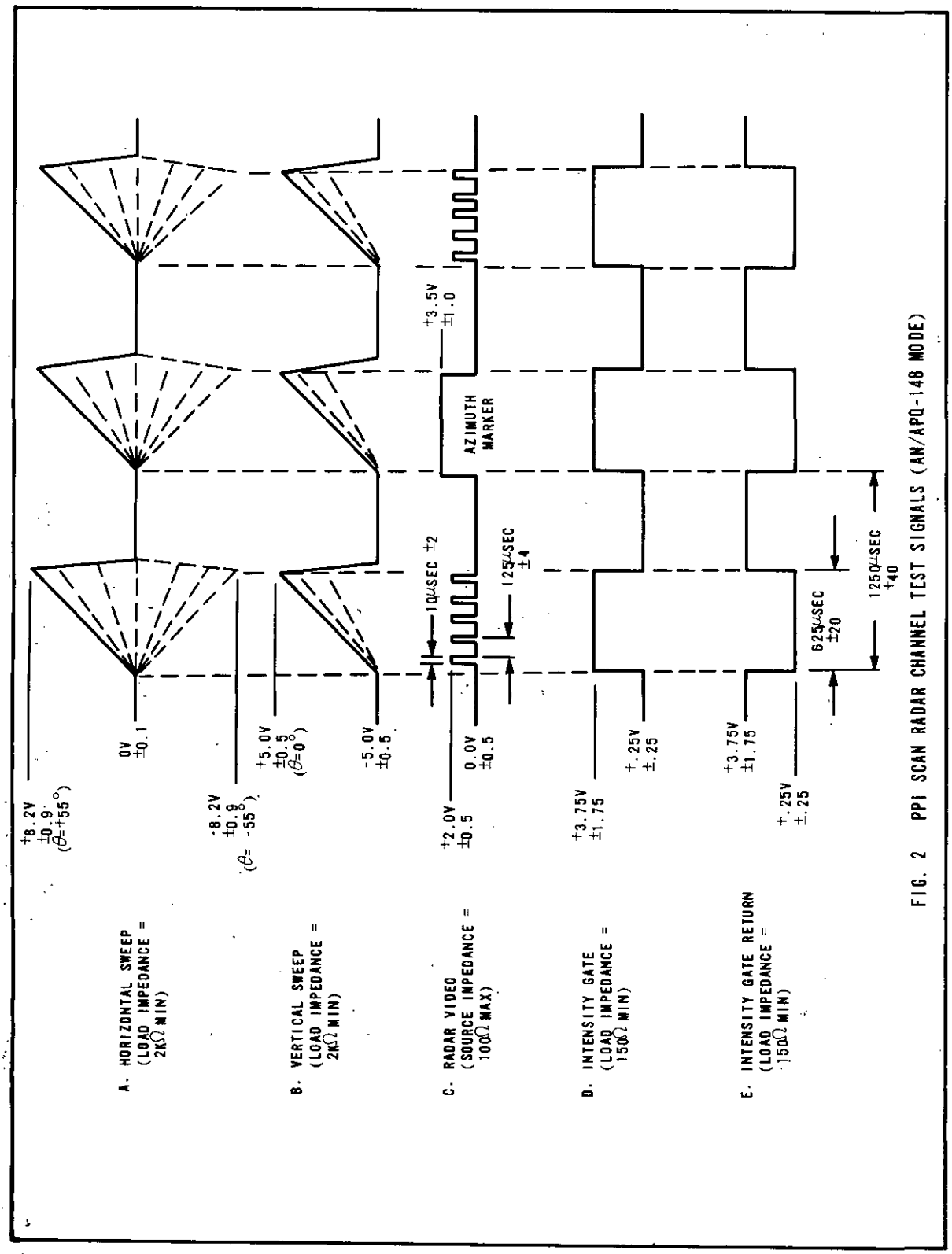


FIG. 2 PPI SCAN RADAR CHANNEL TEST SIGNALS (AN/APQ-148 MODE)

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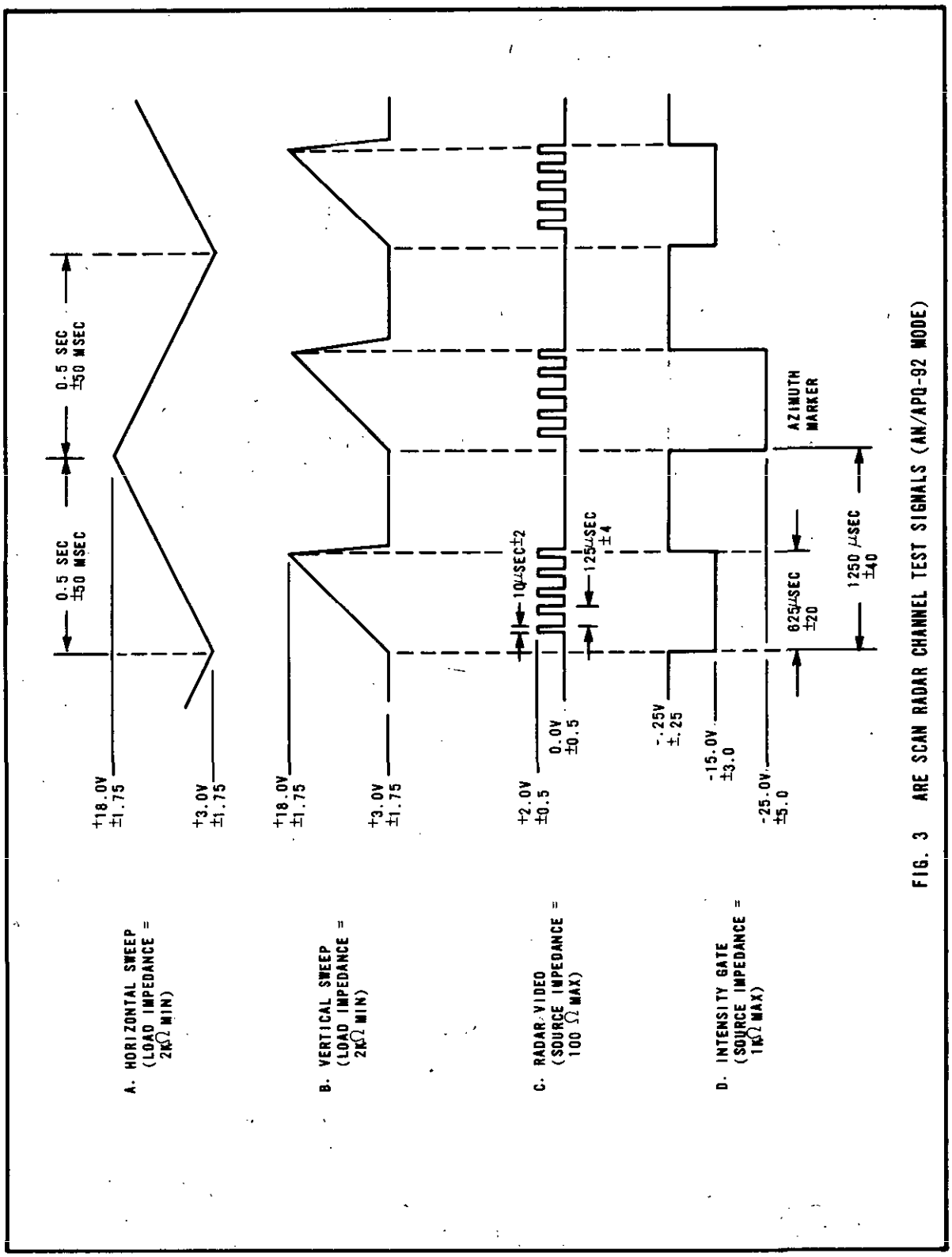


FIG. 3 ARE SCAN RADAR CHANNEL TEST SIGNALS (AN/APQ-92 MODE)

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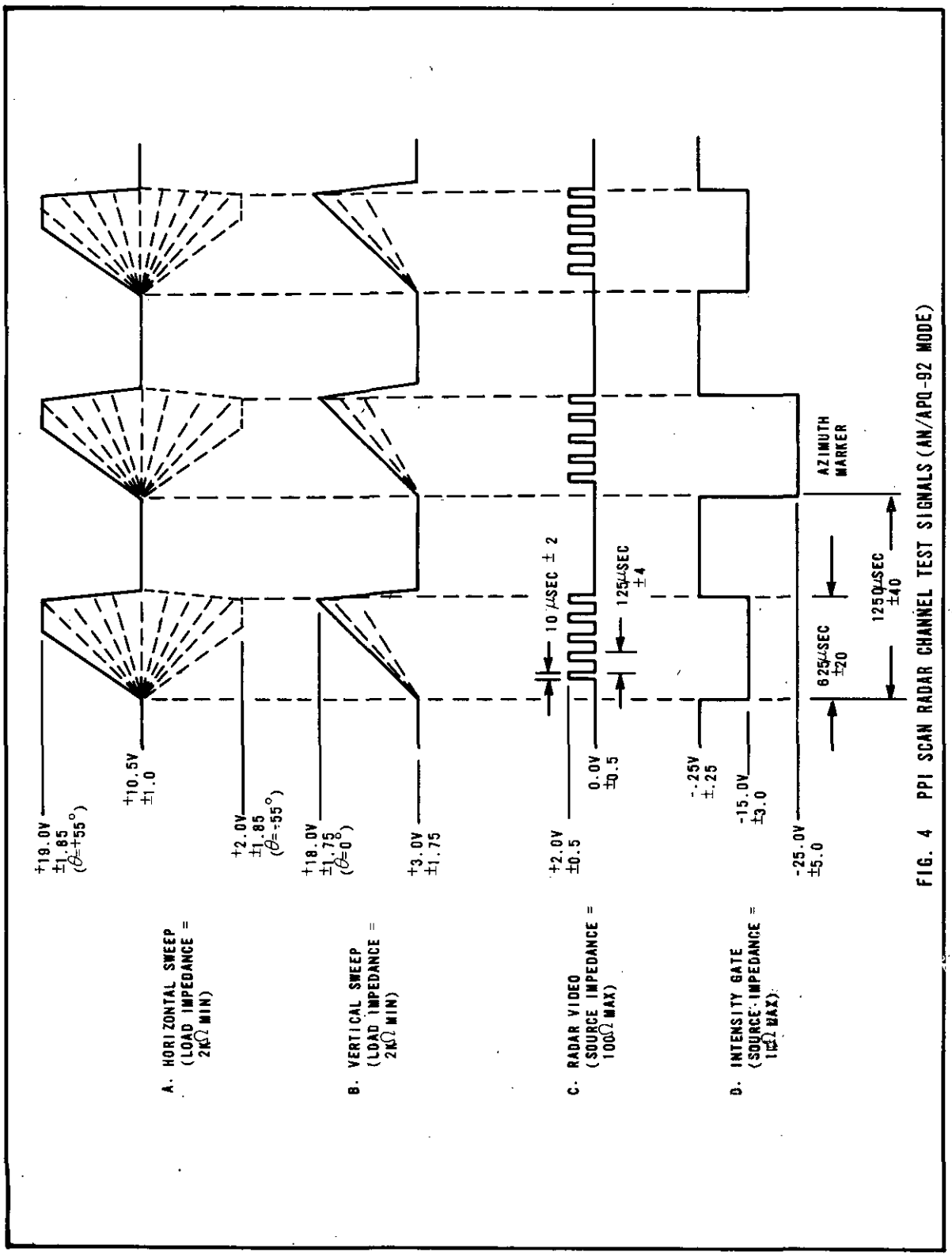


FIG. 4 PPI SCAN RADAR CHANNEL TEST SIGNALS (AN/APQ-92 MODE)

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3.6.1.2 Vertical Sweep output - The output signal shall conform to Waveform B, Figure 1, 2, 3 or 4; depending upon selected mode and scan.

3.6.1.3 Radar Video output - The output signal shall conform to Waveform C, Figures 1, 2, 3 or 4; depending upon selected mode and scan.

3.6.1.4 Intensity Gate output - The output signal shall conform to Waveform D, Figure 1, 2, 3 or 4; depending upon selected mode and scan. Intensity Gate Return signal as required in the AN/APQ-148 mode shall conform to Waveform E, Figures 1 and 2, depending upon selected scan.

3.6.1.5 Audio output - Audio tone test signal shall be an approximate square wave of 1.4 V + 0.4 peak-to-peak amplitude and a frequency of 800 Hz ± 25 (or period of 1250 μsec ± 40). Capability for an audio voice test signal shall also be included in the equipment. The audio output load impedance shall be 600 ohms.

3.6.1.6 Data Gate output - The Data Gate output shall consist of two complementary signals. The true signal shall conform to Waveform A, Figure 5.

3.6.1.7 Data Shift output - The Data Shift output shall consist of two complementary signals. The true signal shall conform to Waveform B, Figure 5.

3.6.1.8 Data output - The Data output shall consist of two complementary signals. The true signal shall conform to Waveforms C, D or E, Figure 5; depending upon selected scan and ARE level (when scan is ARE).

3.6.1.9 ARE 30 output - The ARE 30 output shall consist of two complementary signals. The true signal shall conform to Waveform F, Figure 5.

3.6.1.10 ARE 60 output - The ARE 60 output shall consist of two complementary signals. The true signal shall conform to Waveform G, Figure 5.

3.6.1.11 Attack Command output - The attack command output shall be a 26 VDC ± 2 signal when ON. The attack command output shall be capable of supplying a 25 milliampere maximum load current.

3.6.1.12 5 VAC output - The equipment shall be capable of supplying 5 VAC ± 0.5 at 1.5 amperes to the Mission Recorder.

3.6.1.13 TV Video output - Provisions for an optional TV video output shall be included as part of the equipment.

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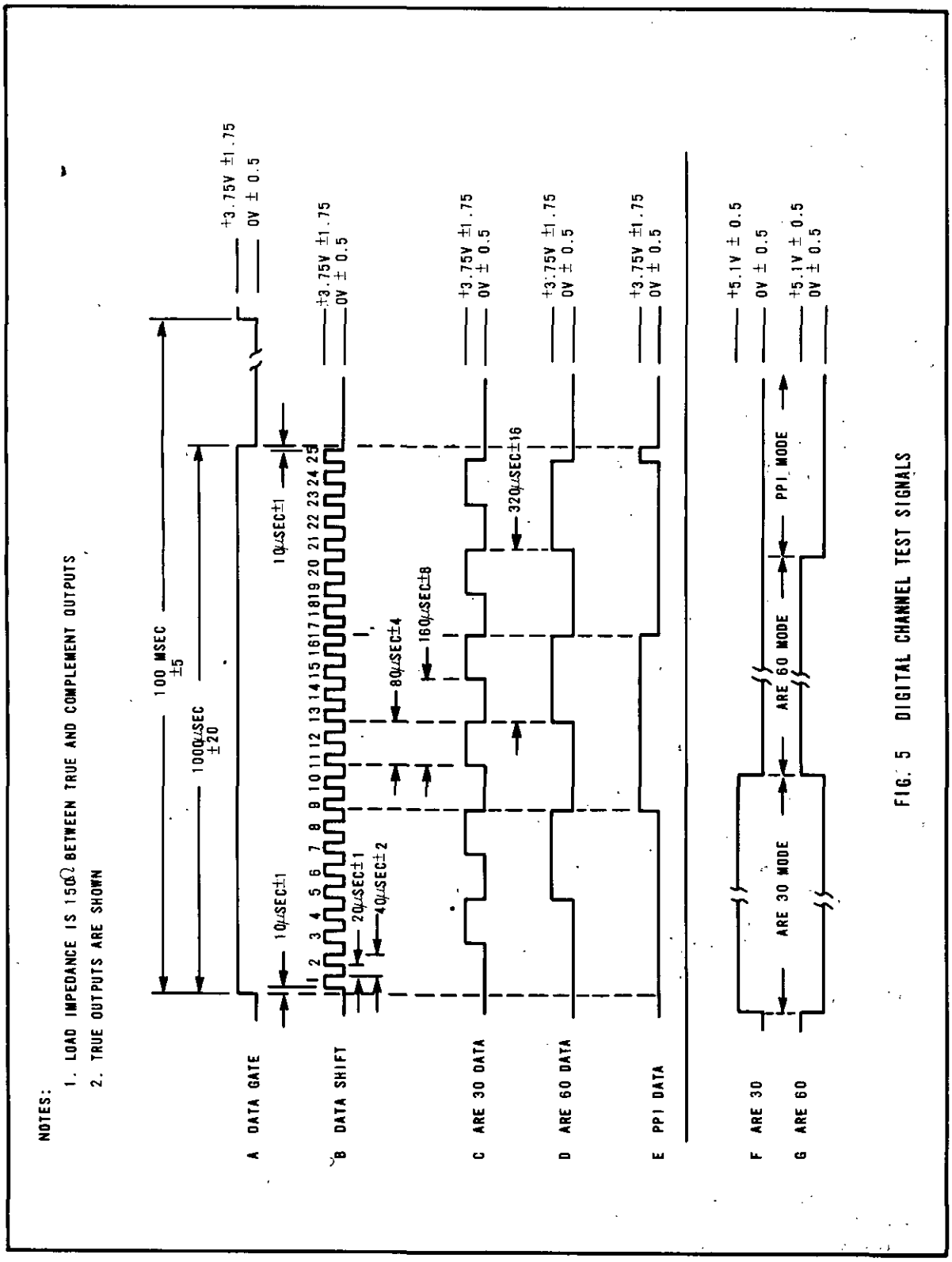


FIG. 5 DIGITAL CHANNEL TEST SIGNALS

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3.6.2 Input signals - The equipment shall accept the input signals described herein from the Mission Recorder. Timing of input signals from the Mission Recorder are as shown in referenced figures.

3.6.2.1 Horizontal Sweep input - The Horizontal Sweep input signal is as shown by Waveform A, Figure 6 or 7; depending upon recorded scan.

3.6.2.2 Vertical Sweep input - The Vertical Sweep input signal is as shown by Waveform B, Figure 6 or 7, depending upon recorded scan.

3.6.2.3 Radar Video input - The equipment shall receive and display radar video signals from the Mission Recorder. Typical radar video signal is shown by Waveform C, Figure 6 or 7; depending upon recorded scan (using bar pattern test tape recorded from the equipment). The video input shall have a 5 MHz minimum bandwidth response and shall be compatible with an AC coupled video signal having the following minimum characteristics:

(1) 1 Volt peak positive video with respect to the video base level.

(2) 0.5 Volt negative pulse with respect to the video base level.

(3) Video source impedance of 75 ohms.

(4) Video signal containing range mark and cursor information.

3.6.2.4 Intensity Gate input - The intensity gate input signal is as shown by Waveform D, Figure 6 or 7. The input shall be compatible with standard TTL (Transistor-Transistor-Logic) positive logic levels. Logic "1" shall allow video to be displayed and Logic "0" shall be for CRT blanking.

3.6.2.5 Audio input - The equipment shall receive and reproduce audio signals from the Mission Recorder. The audio input shall be compatible with an audio signal having the following characteristics:

(1) 1 Volt peak-to-peak amplitude, nominal.

(2) Source impedance of 600 ohms.

(3) Within communications voice frequency range.

3.6.2.6 Data input - The equipment shall receive data signals from the Mission Recorder. Typical data signals are shown in Figure 8 (using test tape recorded from the equipment).

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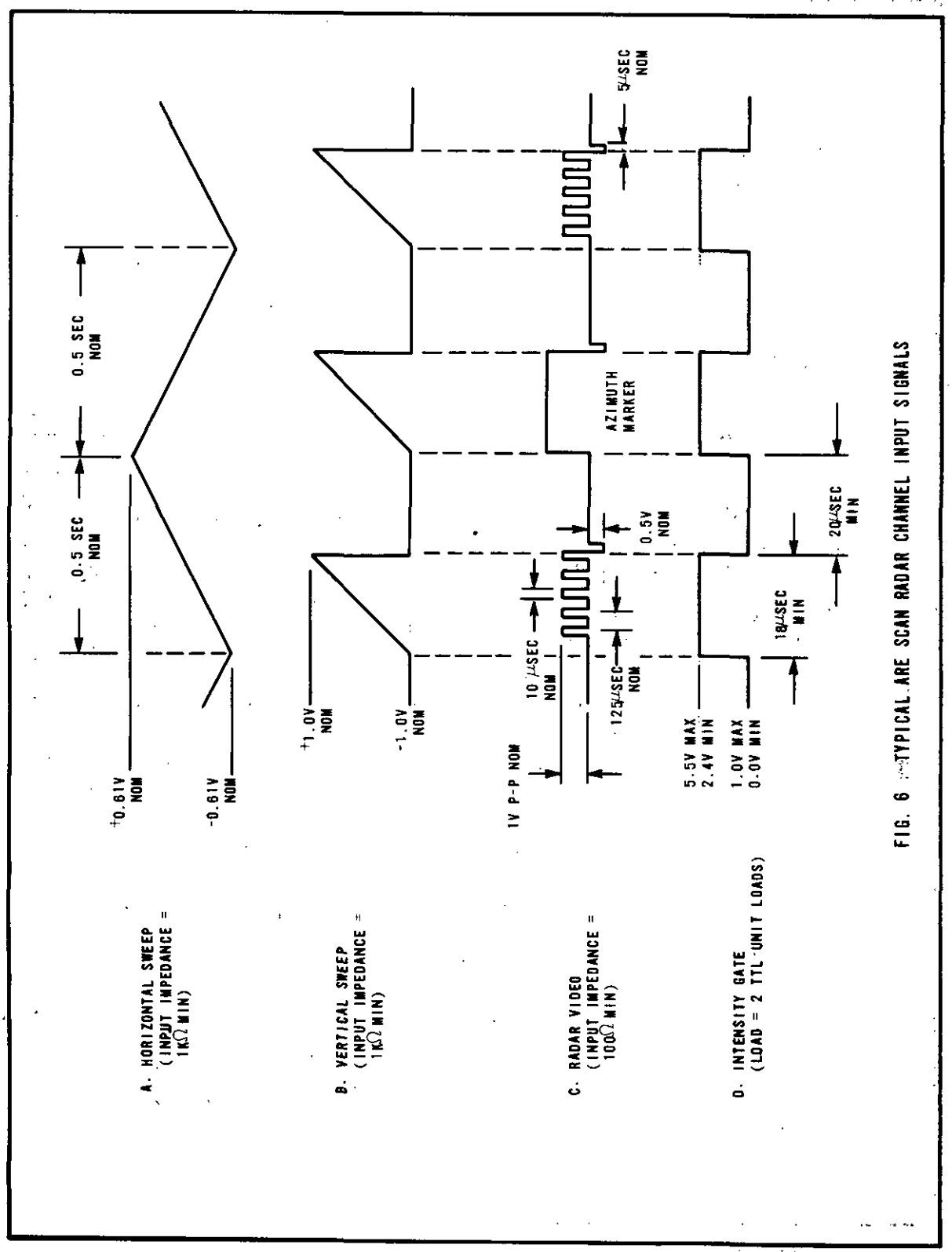


FIG. 6 TYPICAL ARE SCAN RADAR CHANNEL INPUT SIGNALS

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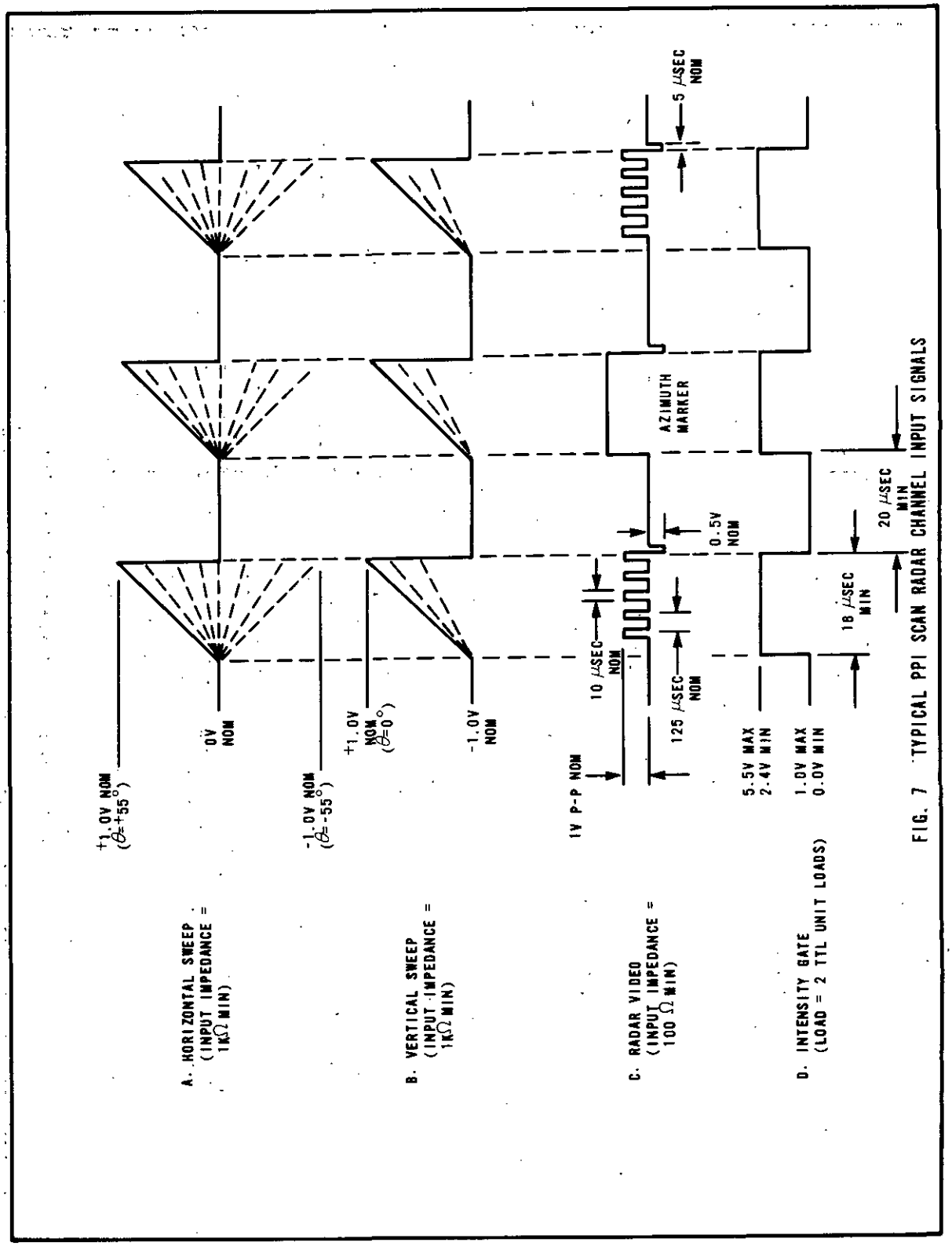


FIG. 7 TYPICAL PPI SCAN RADAR CHANNEL INPUT SIGNALS

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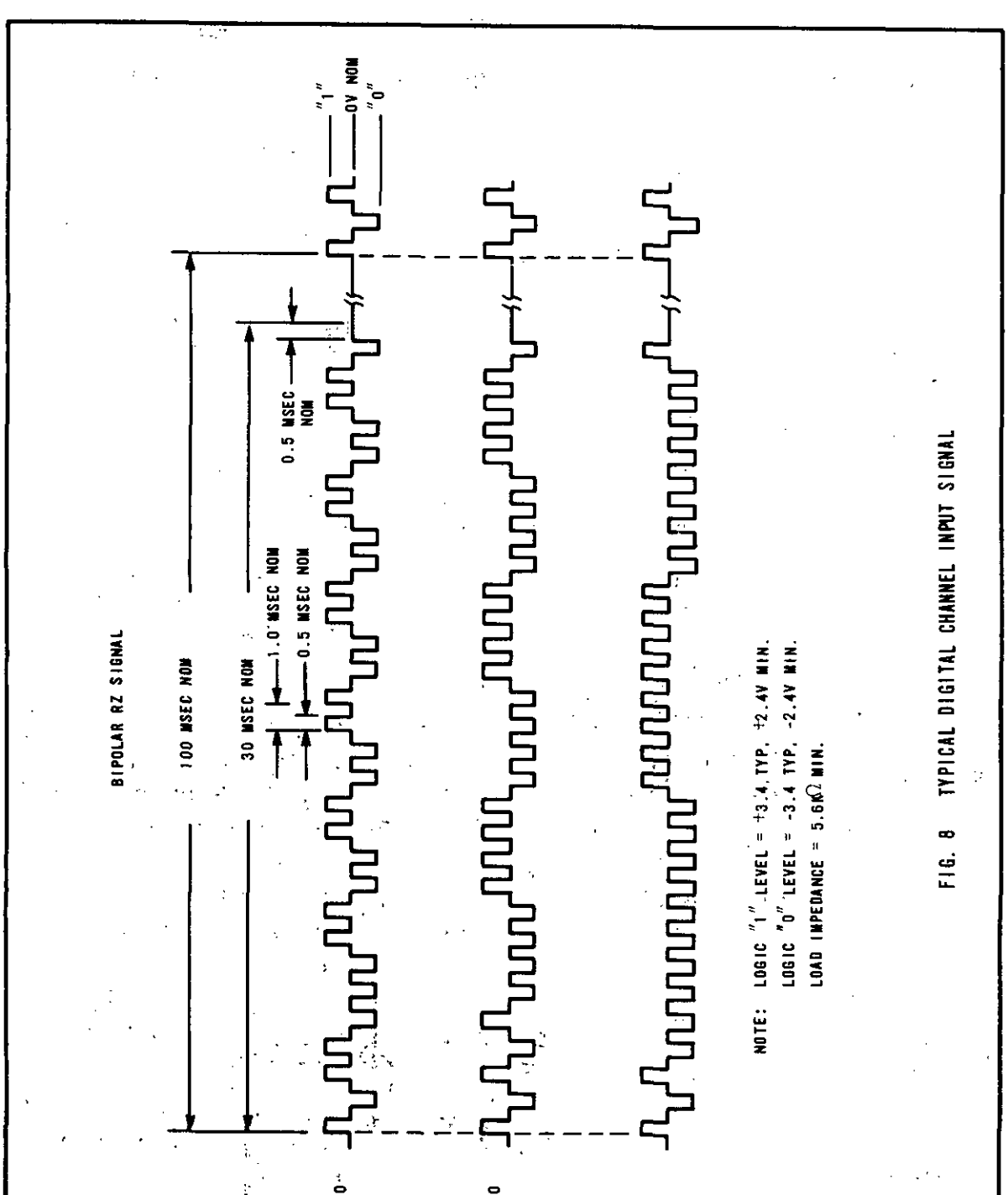


FIG. 8 TYPICAL DIGITAL CHANNEL INPUT SIGNAL

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3.6.2.7 TV Video input - Provisions for an optional TV video input shall be included as part of the equipment.

3.6.3 Fault Detection - The equipment shall incorporate fault detection circuitry.

3.6.3.1 Purpose - The fault detection circuitry shall be for aiding in the isolation of faulty signals reproduced by the Mission Recorder and as a confidence check of the signals produced by the equipment.

3.6.3.2 Operation - The fault detection circuitry shall monitor signals routed to the Radar Indicator. The Radar Indicator signals monitored shall be input signals from the Mission Recorder when the equipment FUNCTION switch is in the REPRODUCE position. The Radar Indicator signals monitored shall be internally generated signals when the equipment FUNCTION switch is in the SELF TEST or RECORD positions.

3.6.3.3 Fault Indication - Indicator lamps shall light to indicate a fault in the following monitored signals:

(1) Radar Video

(2) Horizontal Sweep

(3) Vertical Sweep

(4) Intensity Gate

A fault lamp shall be provided to indicate a fault in the Radar Indicator when the equipment FUNCTION switch is in the SELF TEST position. A FLIR Video fault lamp shall be provided as part of the equipment for future use by the procuring activity.

3.6.4 Radar Display - The equipment shall be capable of displaying either PPI or B-Scan radar presentations. The display shall have the following characteristics:

3.6.4.1 Indicator Tube - The indicator tube shall be a seven inch cathode ray tube, type 7M111P14M, electromagnetically deflected and electrostatically focused. The phosphor shall be P14 type.

3.6.4.2 Brightness - The maximum display tube brightness shall not be less than 5 foot-lamberts when measured on the face of the indicator tube.

3.6.4.3 Spot Size - The indicator tube spot size at a brightness level of 5 foot-lamberts or less shall not exceed 0.007 inches at any point on the face of the tube.

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3.6.4.4 Magnetic Shield - A high-permeability magnetic shield shall be provided to protect the display tube from external fields.

3.6.4.5 Linearity - Display non-linearity introduced by the indicator shall not exceed 1.5 percent.

3.6.4.6 Tonal Renditions - The dynamic range of display brightness shall not be less than 15 dB. This is equivalent to a tonal rendition of five grey shades. Suitable gamma correction shall be provided, if necessary, to meet this requirement.

3.6.4.7 Protective Circuitry - Circuitry shall be provided to protect the cathode ray tube from damage due to a loss of sweep voltage.

3.6.5 Electrical power source - The equipment shall give specified performance when energized from the following power sources having characteristics and limits as defined in the applicable document. The power required shall not exceed the specified amount.

(1) AC Power (Single Phase) 115 V, 60 Hz, Type I (MIL-STD-761), 550 VA

(2) AC Power (Single Phase) 115 V, 400 Hz, Category B (MIL-STD-704), 400 VA

(3) DC Power, 28V, Category B (MIL-STD-704), 290 Watts

3.6.5.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

3.6.6 Warm-up time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

3.7 Detail requirements -

3.7.1 Power Supply Group OP-87/USM-387 - The Power Supply Group shall consist of the following Weapons Replaceable Assemblies (WRA's):

(1) Mounting Base, Electrical Equipment MT-4629/USM-387

(2) Power Supply PP-6743/UPQ-5

3.7.1.1 Mounting Base, Electrical Equipment MT-4629/USM-387 - The Mounting Base shall meet the following requirements:

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3.7.1.1.1 Function - The Mounting Base shall be a mechanical facility to hold the Power Supply, RFI Filter Assembly and interconnecting cables.

3.7.1.1.2 Form factor - The Mounting Base shall not exceed 19 inches wide, 28 inches deep and 9 inches high, excluding mounting ears.

3.7.1.1.3 Weight - The weight of the Mounting Base shall not exceed 40 pounds.

3.7.1.1.4 Content - The Mounting Base shall contain the following items:

(1) RFI Filter Assembly - The RFI Filter Assembly shall provide RFI filtering of each side of the 115 VAC, 60 Hz input power line. The RFI Filter Assembly shall weigh 12 pounds, maximum and size shall not exceed 6 inches wide, 12 inches deep and 3 inches high.

(2) Cable Assemblies W1 and W2 - Cable assemblies W1 and W2 shall provide interconnection between the electrical power source and the RFI Filter Assembly (W1) and between the RFI Filter Assembly and the Power Supply (W2).

3.7.1.2 Power Supply PP-6743/UPQ-5 - The Power Supply shall meet the following requirements:

3.7.1.2.1 Function - The Power Supply shall serve as a distribution point for 115 VAC to the Oscilloscope and Signal Generator. It shall also contain the +32 VDC and -32 VDC power supplies and the main power switch.

3.7.1.2.2 Form factor - The Power Supply shall be rack-mounted and shall not exceed 8-3/4 inches high, 19 inches wide and 20 inches deep.

3.7.1.2.3 Weight - The weight of the Power Supply shall not exceed 119 pounds.

3.7.1.2.4 Materials, parts, processes, design and construction - The materials, parts, processes, design and construction of the Power Supply shall be in accordance with MIL-E-16400.

3.7.1.2.5 Content - The Power Supply shall contain the following subassemblies:

(1) Chassis Assy, Power Supply - The chassis assembly shall provide mounting facilities and interconnecting wiring for the +32 VDC and -32 VDC power supplies, front panel controls and external cables.

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(2) Power Supply Subassy, +32 VDC - A +32 VDC supply capable of supplying 6.0 amperes.

(3) Power Supply Subassy, -32 VDC - A -32 VDC supply capable of supplying 6.0 amperes.

3.7.1.2.6 Electrical Connections - Connections to external circuits shall be provided as follows (see figure 9):

Reference Designation	Receptacle Type	Function
J1	L22TE19PONA* (or an approved equivalent)	Input power
J2	MS3120E-12-3S	Not used
J3	MS3120E-12-3S	Power to the Oscilloscope (2A2)
J4	MS3120E-12-3S	Not used
J5	MS3120E-22-41S	Power to Signal Generator (2A3)

3.7.1.2.7 Test Points - Front panel test points shall be provided to measure the output voltage of the +32 and -32 volt DC supplies and the AC voltage.

3.7.1.2.8 Circuit Breakers - A circuit breaker shall be provided on the input of each power supply subassembly and for the equipment.

3.7.2 Test Set Group, Recorder OQ-184/USM-387 - The Recorder Test Set Group shall consist of the following WRA's:

(1) Indicator, Radar IP-1172/USM-387

(2) Oscilloscope (GFE) AN/USM-296

(3) Generator, Signal SG-1062/USM-387

(4) Cabinet, Electrical Equipment CY-7389/USM-387

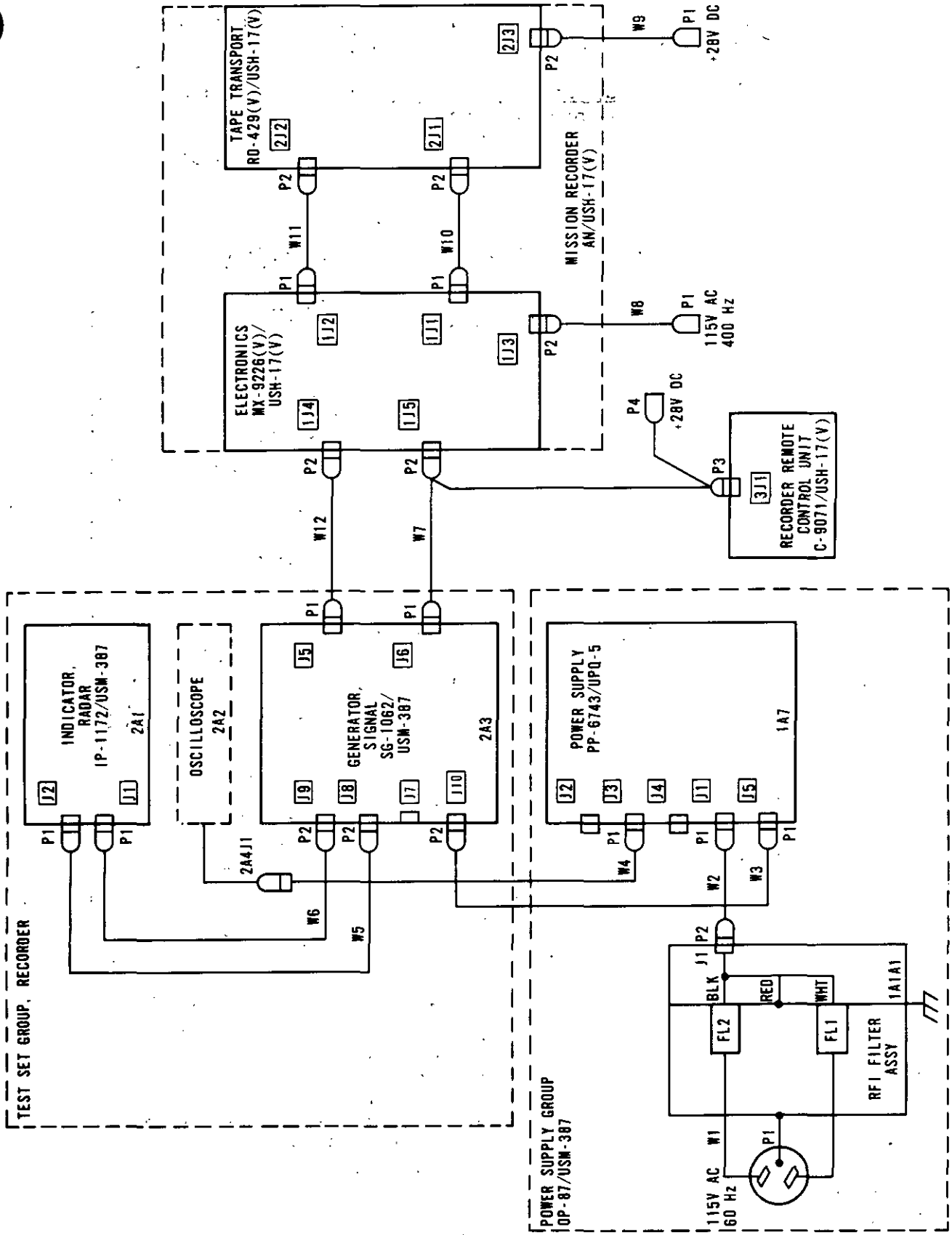
3.7.2.1 Indicator, Radar IP-1172/USM-387 - The Radar Indicator shall meet the following requirements:

3.7.2.1.1 Function - The Radar Indicator shall provide the radar display and shall also contain the audio amplifier and speaker.

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*Burndy Corp., part number.

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3.7.2.1.2 Form factor - The Radar Indicator shall be rack-mounted and shall not exceed 8-3/4 inches high, 19 inches wide and 20 inches deep, excluding handles and control knobs.

3.7.2.1.3 Weight - The weight of the Radar Indicator shall not exceed 50 pounds.

3.7.2.1.4 Materials, parts, processes, design and construction - The materials, parts, processes, design and construction of the Radar Indicator shall be in accordance with MIL-E-16400.

3.7.2.1.5 Content - The Radar Indicator shall contain the below listed subassemblies and printed wiring assemblies (PWA's). PWA's shall be 5 inches by 6-1/4 inches with electrical interface terminated into an M21097/14-07 connector in accordance with MIL-C-21097.

(1) Electron Tube Assy, Cathode Ray - The CRT shall be used as the radar display indicator. The yoke shall provide magnetic deflection of the electron beam.

(2) Divider, High Voltage - This subassembly shall provide a means for measuring the CRT anode voltage. A voltage division of 1000 shall be used.

(3) Power Supply Assy, HV - This power supply shall provide 8500 volts at current up to 500 micro-amperes to the radar display CRT. This subassembly shall be nonrepairable.

(4) Amplifier-Driver, Deflection Coil - This sub-assembly shall provide vertical deflection current to the deflection yoke of the radar display CRT. It shall be a linear voltage-to-current converter.

(5) Amplifier-Driver, Deflection Coil - This sub-assembly shall be identical to (4) above, except it shall provide the horizontal deflection current.

(6) Circuit Card Assembly, Video Amplifier - This PWA shall provide amplification and dc restoration of the input video signal.

(7) Circuit Card Assembly, Unblanking and Phosphor Protect - This PWA shall provide CRT unblanking from the intensity gate signal and protect the CRT in the event of sweep or CRT cathode bias loss.

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(8) Circuit Card Assembly, Audio Amplifier - This PWA shall be a power amplifier for the audio.

(9) Circuit Card Assembly, Fault Sensing - This PWA shall monitor selected circuit outputs during the SELF TEST mode and shall energize a fault indicator light in the event that any of the selected outputs is absent.

(10) Circuit Card Assembly, Extender - This PWA shall provide electrical connection between an extended assembly and its chassis connector.

(11) Bias Supply - This subassembly shall provide focus voltage and screen voltage for the CRT and the voltage for (7) above.

(12) Chassis Assembly, Indicator, Radar - This chassis assembly shall provide the mounting facilities and interconnecting wiring to interface the above sub-assemblies with the front panel controls and external cables.

3.7.2.1.6 Controls - The Radar Indicator shall have the following front panel controls:

Control	Function
CONTRAST	Adjust display contrast
BRIGHTNESS	Adjust display brightness
VERTICAL	Position display vertically
HORIZONTAL	Position display horizontally
AUDIO LEVEL	Adjust audio level

3.7.2.1.7 Electrical Connections - Connections to external circuits shall be provided as follows (see figure 9).

Reference Designation	Receptacle Type	Function
J1	MS3120E-22-41P	Power from Signal Generator
J2	L22TE95PONA* (or an approved equivalent)	Signal from Signal Generator

*Burndy Corp., Part Number.

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3.7.2.2 Oscilloscope AN/USM-296 - The Oscilloscope shall be furnished by the procuring activity and shall meet the following requirements:

3.7.2.2.1 Function - The Oscilloscope shall be used with the equipment during maintenance of the Mission Recorder and as a maintenance aid for the equipment.

3.7.2.2.2 Form factor - The Oscilloscope shall be rack-mounted and shall not exceed 5-1/4 inches high, 19 inches wide and 20 inches deep, excluding handles and knobs.

3.7.2.2.3 Weight - The weight of the Oscilloscope shall not exceed 40 pounds.

3.7.2.3 Generator, Signal SG-1062/USM-387 - The Signal Generator shall meet the following requirements:

3.7.2.3.1 Function - The Signal Generator shall generate the test signals for the Mission Recorder and route generated or reproduce signals to the Radar Indicator.

3.7.2.3.2 Form factor - The Signal Generator shall be rack-mounted and shall not exceed 7 inches high, 19 inches wide and 20 inches deep, excluding handles and knobs.

3.7.2.3.3 Weight - The weight of the Signal Generator shall not exceed 54 pounds.

3.7.2.3.4 Content - The Signal Generator shall contain the below listed subassemblies and printed wiring assemblies (PWA's). PWA's shall be 4-3/4 inches by 6-3/4 inches with electrical interface terminated into an M21097/14-13 connector in accordance with MIL-C-21097.

(1) Circuit Card Assembly, Sine/Cosine Generator - This PWA shall generate the envelope of the horizontal and vertical sweep waveforms for a PPI scan radar display.

(2) Circuit Card Assembly, Sweep Generator - This PWA shall generate the horizontal and vertical sweep waveforms for both PPI and ARE scan radar displays.

(3) Circuit Card Assembly, Digital Generator - This PWA shall generate the digital data signals to be sent to the Mission Recorder.

(4) Circuit Card Assembly, Digital Output - This PWA shall buffer the digital data signals and AN/APQ-148 intensity gate signal to levels acceptable by the Mission Recorder. This assembly shall also contain a 3 MHz crystal-controlled oscillator counted down to provide 40 kHz and 50 kHz for the other assemblies.

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(5) Circuit Card Assembly, Audio/Video Generator- This PWA shall generate the video and the audio test signals and buffer the audio signal from the external microphone.

(6) Circuit Card Assembly, Sweep Converter - This PWA shall convert the horizontal sweep, vertical sweep and intensity gate signals from AN/APQ-148 levels to levels of the AN/APQ-92 radar.

(7) Circuit Card Assembly, Fault Sensing - This PWA shall detect the presence of radar video, horizontal sweep, vertical sweep and intensity gate signals going to the Radar Indicator and shall cause the associated fault lamp to be illuminated if any signal is absent.

(8) Circuit Card Assembly, Extender - This PWA shall provide electrical connection between an extended assembly and its chassis connector.

(9) Power Supply Subassembly, +6 VDC - A 6 VDC supply capable of supplying 3.0 amperes.

(10) Power Supply Subassembly, $\pm$15 VDC - A $\pm$15 VDC supply capable of supplying 0.5 amperes.

(11) Power Supply Subassembly, +450 VDC - A 450 VDC supply capable of supplying 0.1 amperes.

(12) Overvoltage Protector Subassembly, +6V - Provide overvoltage protection on the output of the 6 VDC supply.

(13) Chassis Assembly, Generator, Signal - This chassis assembly shall provide mounting facilities and interconnecting wiring between the above listed subassemblies, the front panel controls and external cables.

3.7.2.3.5 Controls - The Signal Generator shall have the following front panel controls:

Control	Function
FUNCTION Switch	Selects between SELF TEST, RECORD and REPRODUCE function of the equipment.
RADAR SCAN Switch	Selects either PPI scan or ARE scan radar channel test signals.
ARE LEVEL Switch	Selects either ARE 30 or ARE 60 digital channel test signals.

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Control	Function
LAMP TEST Switch	Used to verify operation of the front panel lamps.
ATTACK COMMAND Switch	Enables the attack command signal.
RADAR MODE Switch (located inside chassis)	Changes test signals to simulate either AN/APQ-92 Radar or AN/APQ-148 Radar.

3.7.2.3.6 Electrical connections - Connections to external circuits shall be provided as follows (see figure 9):

Reference Designation	Receptacle Type	Function
J1	M39012/21-0002	Reproduce Digital Data output
J2	M39012/21-0002	Reproduce Digital Data Sync output
J3	M39012/21-0002	Reproduce Test Sync output
J4	M641/10-01	External Audio Input jack
J5	MS24264G20B28PN	Reproduce Signals from Mission Recorder
J6	MS24264G24B57SN	Record Signals to Mission Recorder
J7	M39012/21-0002	TV Monitor Video output
J8	L22TE95SONA* (or an approved equivalent)	Signals to Radar Indicator

*Burndy Corp., Part Number.

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Reference Designation	Receptacle Type	Function
J9	MS3120E-22-41SX	Power to Radar Indicator
J10	MS3120E-22-41P	Power from Power Supply

3.7.2.3.7 Test Points - The Signal Generator shall have the front panel test points shown on Table I.

TABLE I. FRONT PANEL TEST POINTS

REFERENCE DESIGNATION	DESCRIPTION
TP1	RECORD - VIDEO SIGNAL
TP2	RECORD - H SWP SIGNAL
TP3	RECORD - V SWP SIGNAL
TP4	RECORD - INT GATE SIGNAL
TP5	RECORD - DATA SIGNAL
TP6	RECORD - SHIFT SIGNAL
TP7	RECORD - GATE SIGNAL
TP8	RECORD - ARE 30 SIGNAL
TP9	RECORD - ARE 60 SIGNAL
TP10	RECORD - AUDIO SIGNAL
TP11	RECORD - FLIR SIGNAL
TP12	REPRODUCE - VIDEO SIGNAL
TP13	REPRODUCE - H SWP SIGNAL
TP14	REPRODUCE - V SWP SIGNAL
TP15	REPRODUCE - INT GATE SIGNAL
TP16	REPRODUCE - AUDIO SIGNAL
TP17	REPRODUCE - FLIR SIGNAL
TP18	ATTACK COMMAND SIGNAL

3.7.2.4 Cabinet, Electrical Equipment CY-7389/USM-387 - The Cabinet shall meet the following requirements:

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3.7.2.4.1 Function - The Cabinet shall provide an enclosure, mounting facilities and interconnecting cabling for the Signal Generator, Radar Indicator and Oscilloscope and have space allotted for a rack-mounted TV Monitor.

3.7.2.4.2 Form factor - The Cabinet shall not exceed 24 inches wide, 28 inches deep and 38 inches high, excluding lifting eyes.

3.7.2.4.3 Weight - The weight of the Cabinet shall not exceed 165 pounds.

3.7.2.4.4 Content - The Cabinet shall contain the following:

(1) Cable Assemblies - The Cabinet shall contain the following cable assemblies which interconnect the Weapons Replaceable Assemblies (WRA's) of the equipment: W3, W4, W5 and W6 (see figure 9).

3.7.3 Test Stand, Recorder MT-4628/USM-387 - The Test Stand shall meet the following requirements:

3.7.3.1 Function - The Test Stand shall be a facility to hold the Mission Recorder and provide interconnecting cables for testing of the Mission Recorder.

3.7.3.2 Weight - The weight of the Test Stand shall not exceed 27 pounds.

3.7.3.3 Content - The Test Stand shall contain the following:

(1) Mounting Fixture, Recorder - The Recorder Mounting Fixture shall be a mechanical facility used to secure the Mission Recorder WRA's to the shop bench. Its size shall not exceed 6 inches high, 37 inches wide and 24 inches deep.

(2) Cable Assemblies - The Test Stand shall contain the following cable assemblies which provide for interconnection between the Mission Recorder Transport and Electronics WRA's, test signal and reproduce signal interconnection between the equipment and the Mission Recorder WRA's, and power connection for the +28 VDC and the 115 V, 400 Hz shop power to the Mission Recorder WRA's: W7, W8, W9, W10, W11 and W12 (see figure 9).

3.7.4 Tape, Recorder Test - The Tape shall meet the following requirements:

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3.7.4.1 Function - The Tape shall provide pre-recorded radar, TV, digital and audio signals used to test the Mission Recorder reproduce electronics.

3.8 Finish - Finishes shall be in accordance with MIL-T-21200.

3.9 Interchangeability - Interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.10 Workmanship - Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

First article (preproduction) tests

Initial production tests

Acceptance tests

4.2 First article (preproduction) tests - First article (preproduction) tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 First article (preproduction) test data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Scope of tests - The tests shall include all tests deemed necessary, by the procuring activity, to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Within the limits specified in MIL-T-21200, first article tests shall include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

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4.2.3 First article (preproduction) approval - Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The test sample shall not be considered as one of the equipments under the contract.

4.2.4 Production equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved test sample. Each equipment shall be capable of successfully passing the same tests as imposed on the test sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production tests - When specified by the procuring activity, one of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The test sample shall not be selected for this test.

4.3.1 Scope of tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial production sample approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of initial production test sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1)	Individual tests	(4.4.1)
(2)	Reliability assurance tests	(4.4.2)
(3)	Special tests	(4.4.3)

4.4.1 Individual tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1)	Examination of product	(4.4.1.1)
(2)	Operational test	(4.4.1.2)
(3)	Manufacturing run in test	(4.4.1.3, when specified)

4.4.1.1 Examination of product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing run-in-test - When specified by the procuring activity, each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operating within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Reliability assurance tests - When specified by the procuring activity, reliability assurance tests shall be conducted using MIL-STD-781.

4.4.2.1 Test details - Test details such as the length of the test cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.4.3 Special tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the reliability assurance tests.

4.4.3.1 Special test schedule - Selection of equipments for special tests shall be made as follows:

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(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in failure tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

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4.6 Reconditioning of tested equipment - Equipment which has been subjected to acceptance tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.7 Presubmission testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

5.2 Marking - In addition to any special marking by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use - The equipment shall be used to generate test signals for, and for monitoring reproduced signals of, the Mission Recorder Set AN/USH-17(V). As such, the design shall ensure good reliability and maintainability in a fleet environment. It is intended that the equipment shall be used and maintained by ground support avionics personnel.

6.2 Performance objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.5 Ordering data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Samples required for parts and material approval (See 3.2.3).

(3) Reliability requirements (See 3.4.1).

(4) Maintainability requirements (See 3.4.2).

(5) Initial production test requirements (See 4.3).

(6) Manufacturing run-in test requirements (See 4.4.1.3).

(7) Reliability assurance test requirements (See 4.4.2).

(8) Selection of applicable levels of packaging and packing (See 5.1).

6.6 Type designation - The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.5. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterist "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated equipment - The equipment shall operate with the following associated equipment:

Multimeter, VOM, AN/USM-311 (FSN 2Z6625-160-1301)

6.9 Specification cognizance - This specification is under the cognizance of AIR-53421B.

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MIL-T-81978(AS) TEST SET, MISSION RECORDER AN/USM-387

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

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